Karunadu Technologies Private Limited

Internship Project

Project Title

Industrial–Residential Air Quality Classification using Machine Learning & Django

Introduction

Air pollution is a major global concern and has a significant impact on human health and the environment. Pollution levels vary significantly between **industrial areas and residential areas** due to industrial emissions, traffic, and urban activities.

In this assignment, students will develop a **Machine Learning-based web application** to classify whether a given air quality measurement belongs to an **Industrial area or Residential area** based on pollutant levels.

Dataset

The dataset contains various attributes such as:

CO – Carbon Monoxide
NO_2 – Nitrogen Dioxide
SO_2 – Sulfur Dioxide
O_3 – Ozone
PM2.5 – Fine particulate matter
PM10 – Coarse particulate matter

Note:Interns will build and compare multiple classification models to predict the output label.

Objective

- Apply Machine Learning classification algorithms
- Understand data preprocessing and feature engineering
- Build a web application using Django
- Compare performance of different algorithms

Backend	Frontend	Expected Output
Python	HTML	➢ User login system
Django	CSS / Bootstrap	➢ Data input form
Framework	Django Templates	➢ Disease prediction output
Scikit-learn		➢ Accuracy comparison of models

Tasks

Interns must implement **at least 4 classification algorithms** from the list below:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes

Tasks

Web Application (Django)

Interns must build a **full-stack web application** using Django.

Login Module

- User Registration
- Login Authentication
- Session Management

Input Form Page

Build Django UI with input form of all Inputs

Prediction Module

Add a **“Predict” button**

On submission:

- Load trained ML models
- Predict output label
- Display result on UI

Results Display

- Predicted Output
- Accuracy of each algorithm
- Best performing algorithm

Important Questions

- What is classification in Machine Learning? How is it different from regression?
- Explain the importance of the output in this dataset.
- What is feature selection? Why is it important in this project?
- Explain the difference between supervised and unsupervised learning.
- What is overfitting and underfitting? How can it be avoided?
- What is the role of training and testing datasets?
- Explain how Logistic Regression works for classification.
- How does a Decision Tree decide splits?
- What is the advantage of Random Forest over Decision Tree?
- Explain how KNN works and its limitations.
- What is the concept behind SVM (Support Vector Machine)?
- What is a Confusion Matrix? Interpret it for your model.
- What preprocessing steps did you apply to the dataset?
- How did you handle missing or inconsistent data?
- Explain the flow of your Django application.
- How does authentication (login system) work in your project?
- What is the role of views, templates, and models in Django?
- How is the prediction triggered from the frontend?

Submission Requirements

Interns must submit

- ➢ Internship Report
- ➢ Source Code (Django + ML models)
- ➢ Dataset (if modified)
- ➢ Project Report (PDF) including:
  - Problem Statement
  - Methodology
  - Algorithms Used
  - Results & Comparison
  - Screenshots

Contact US

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Location

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Thank You Interns for being a part of Internship Journey

Contact Karunadu Technologies Private Limited For Internship